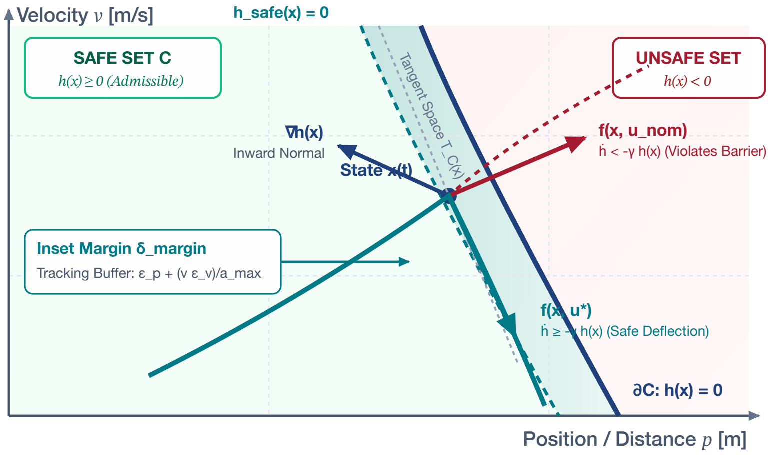
CONTROL BARRIER FUNCTIONS (CBF) & MINIMAL INTERVENTION

Forward Invariance in State Space and Orthogonal Quadratic Program Projection in Control Space

(a) State Space Invariance: Safe Set C & Tangential Filtering



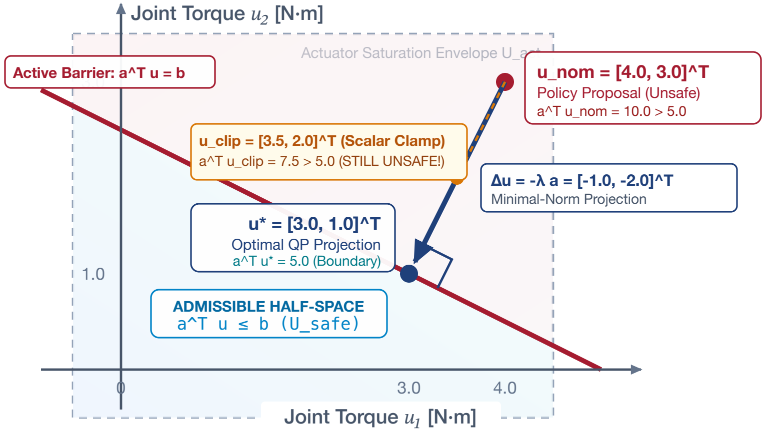
FORWARD INVARIANCE CERTIFICATE

`ḣ(x,u) = L_f h(x) + L_g h(x) u ≥ −γ h(x)`

- **Invariant Safe Set:** Zero-superlevel set $C = \{x \in X \mid h(x) \geq 0\}$ is forward invariant.
- **Nagumo's Theorem:** Control vector field $f(x, u^*)$ remains inside tangent cone $T\_C(x)$.
- **Tracking Inset Margin:** `h_safe(x) = h(x) − δ_margin` buffers tracking error ε_track.
- **Kinematic Inset Formula:** `δ_margin = ε_p + (v ε_v)/a_max` (accounts for loop delay).

Result: Policy commands are filtered tangentially without task abortion.

(b) Control Space: Orthogonal QP Projection onto Admissible Half-Space



QUADRATIC PROGRAM SAFETY FILTER

`u* = argmin_{u ∈ U} 1/2 ||u − u_nom||² s.t. a^T u ≤ b`

- **Closed-Form Analytical Solution:** `u* = u_nom − (max(0, a^T u_nom − b) / ||a||²) a`
- **Zero Intervention Interior:** If $a^T u\_nom \leq b$, then $u^* = u\_nom$ (unmodified policy intent).
- **Coupled Multi-Axis Dynamics:** Preserves Cartesian wrench alignment vs scalar clipping.
- **Policy Health Telemetry:** Rolling norm $\Sigma \, ||u^* - u\_nom|| \, \Delta t$ tracks upstream model drift.

Guarantee: Microsecond deterministic QP projection on embedded MCU SRAM.